

A Charge Domain SRAM Compute-in-Memory Macro With C-2C Ladder-Based 8-Bit MAC Unit in 22-nm FinFET Process for Edge Inference

Hechen Wang^{ID}, *Member, IEEE*, Renzhi Liu^{ID}, *Member, IEEE*, Richard Dorrance^{ID}, *Senior Member, IEEE*, Deepak Dasalukunte^{ID}, *Member, IEEE*, Dan Lake^{ID}, *Member, IEEE*, and Brent Carlton^{ID}, *Member, IEEE*

Abstract—Compute-in-memory (CiM) is one promising solution to address the memory bottleneck existing in traditional computing architectures. However, the tradeoff between energy efficiency and computing precision plagues most CiM implementations, and the low precision imposes a major limitation on CiM's ability to support practical computational workloads. In this article, a static random access memory (SRAM)-based analog CiM macro is presented with the Intel 22FFL process. By introducing a 1-to-2 ratioed capacitor ladder (C-2C)-based charge domain computing scheme, the proposed CiM prototype chip demonstrates a maximum of 2k multiply-accumulate (MAC) operations in one clock cycle and achieves 32.2-TOPS/W peak power efficiency with 8-bit precision in both input activation and weight while ensuring accurate on-chip matrix–vector multiplications (MVMs) with a computation error less than 0.5%. A 4.0-TOPS/mm² peak area efficiency is attained by adopting a local weight multiplexing scheme with a 9T SRAM cell, which improves the memory density and reduces the need to refresh the weight stored in the SRAM array. A variety of analog impairment factors, including parasitics, mismatch, and noise, were analyzed to guarantee a sufficiently high multibit linearity. The proposed passive analog computing mechanism ensures the computation accuracy over process–voltage–temperature (PVT), with the measured MVM error deviation of less than 1% over PVT variations.

Index Terms—Analog computing, charge domain computing, CMOS, compute-in-memory (CiM), machine learning (ML) accelerator, mixed signal, multiply-accumulate (MAC) operation, neural networks (NNs), static random access memory (SRAM).

I. INTRODUCTION

IN THE era of artificial intelligence (AI), machine learning (ML) and neural networks (NNs) have already become unobtrusive in our daily lives. Over the past several years, a wide range of AI techniques, including image classification, object identification, natural language processing, and facial recognition, have been thoroughly incorporated into many of our day-to-day scenarios [1], [2], [3], [4], [5].

In the current paradigm, most ML applications are processed by centralized computation engines on the clouds, which are

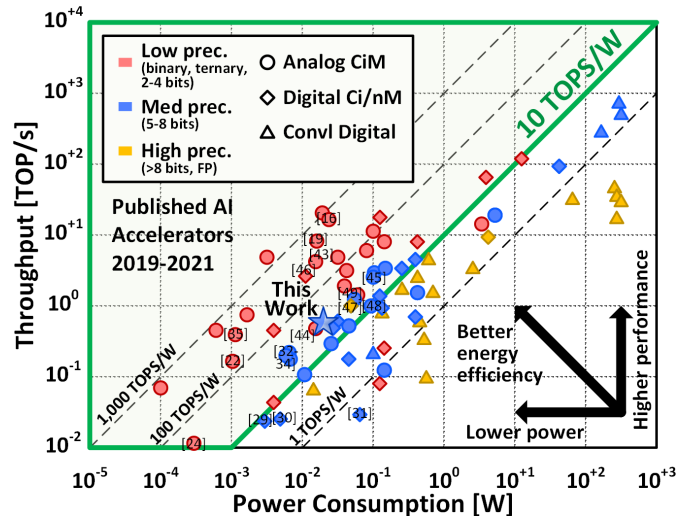


Fig. 1. AI accelerator performance comparison chart of energy efficiency versus weight precision.

normally provided by several giant data center corporations. While we enjoy the benefits of those AI applications, this cloud-based paradigm poses multiple issues, especially in the aspects of latency and privacy. Depending on the network bandwidth, users' location, and the size of data that needs to be transmitted, it may take a noticeable amount of time to get the answer from a data center if the data are originating at an end device. Also, the service can even be interrupted if the network becomes unavailable. Privacy is another major concern since cloud-based services are only offered after the user agrees to grant data centers permission to collect personal data over the network, which can potentially expose users to the risk of data center breaches or eavesdropping during data uploading and receiving.

To address these challenges, edge inference and federated learning with pretrained NNs and local data processing capability reduce or even eliminate the need for transferring data to the cloud and have emerged as the next paradigm [6], [7], [8], [9], [10]. In contrast to data centers, energy efficiency takes precedence over all other considerations for devices at the edge. However, current AI techniques are heavily limited by the high-power dissipation associated with conventional digital computing architectures and techniques. Dominated by highly parallel matrix–vector multiplications (MVMs), NN-based ML inference, which requires frequent data movement between the computing elements and the mem-

Manuscript received 29 August 2022; revised 4 November 2022 and 16 December 2022; accepted 17 December 2022. Date of publication 6 January 2023; date of current version 28 March 2023. This article was approved by Associate Editor Borivoje Nikolić. (Corresponding author: Hechen Wang.)

The authors are with Intel Labs, Intel Corporation, Hillsboro, OR 97124 USA (e-mail: hechen.wang@intel.com).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2022.3232601>.

Digital Object Identifier 10.1109/JSSC.2022.3232601

0018-9200 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

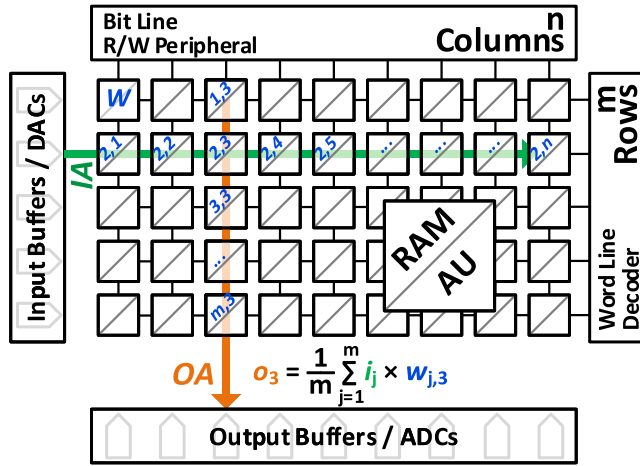


Fig. 2. Basic concept of the analog CiM architecture with AU units merged with RAM cells.

ory units, is facing the so-called “memory wall” or “memory bottleneck” in conventional digital computation architectures. Fig. 1 summarizes the energy efficiency of some recently published application-specific integrated circuit (ASIC) ML accelerators. Traditional digital schemes, as represented by the triangle markers, are still limited to the sub-10-TOPS/W region, making them difficult to deploy in energy-constrained edge devices [11], [12], [13], [14], [15].

Several attempts have been made to solve this dilemma. The concepts of compute-in-memory (CiM) and compute-near-memory (CnM) have been proposed in recent years, which have now garnered significant attention. By bringing arithmetic units (AUs) close to or even inside the memory array, CnM and CiM greatly reduce the energy and performance costs of data movement. Considering a balanced AU-to-memory overhead ratio, analog and mixed-signal-based CiMs (ACiMs) have shown superior energy and area efficiency over their digital counterparts. A basic conceptual architecture of CiM is given in Fig. 2, in which the arithmetic unit merges with random access memory (RAM) cells. However, the majority of those published ACiM implementations, round markers in Fig. 1, can only support low-precision data. This greatly restricts the types of edge inference applications that can be supported and excludes the possibility of edge training. Our target is to leverage the high efficiency of ACiM to reach the region above 10 TOPS/W in Fig. 1 while supporting high data precision with at least 8 bit to realize the edge intelligence.

A multibit static RAM (SRAM)-based charge domain CiM prototype is presented in this article with several key techniques: 1) a 1-to-2 ratioed capacitor ladder (C-2C) in-memory structure for energy and area efficient multibit multiply-accumulate (MAC) operation; 2) a local weight multiplexing scheme to improve memory density and decouple SRAM read/write operations from CiM computation; 3) a customized 9T SRAM cell for better disturbance isolation; and 4) low overhead data conversion solutions and an integrated light-weight controller. The rest of this article is organized as fol-

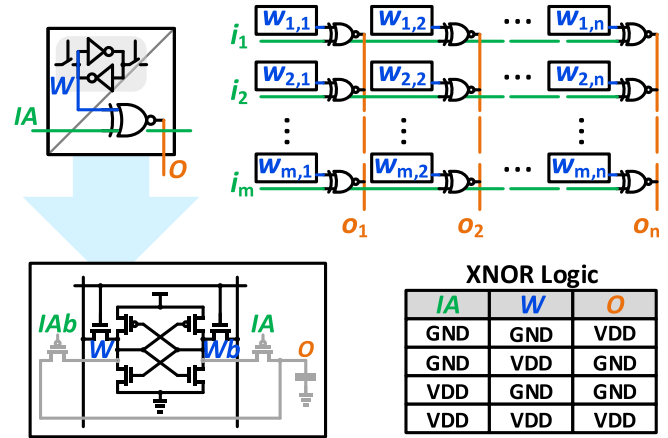


Fig. 3. XNOR-based single-bit CiM solution.

lows. Section II describes the design challenges and tradeoffs of the existing CiMs. The proposed C-2C capacitor ladder-based multibit analog MAC and its operating mechanism are presented in Section III. The corresponding analog computing impairments are analyzed in Section IV. Section V covers the details of the CiM core, peripheral data converter circuits, and upper level system implementations, followed by presenting measurement results in Section VI. Finally, conclusions are drawn in Section VII.

II. MULTIBIT CiM CHALLENGES

Previously reported ACiMs have largely employed XNOR gates for their MAC units [16], [17], [18], [19]. As shown in Fig. 3, by adding only two additional transistors and a capacitor, this CiM approach achieves exceptional energy efficiency with considerably low area overhead when integrated into the conventional 6T SRAM cell. Most of the over 1000-TOPS/W CiM data points are the successors to this scheme. However, xnor logic only provides limited 1-bit precision, leading to a significant degradation in accuracy for most practical NN models. The main task in this area right now is to support multibit weight (W), input activations (IAs), and MAC operations while maintaining reasonable power and area efficiency. A variety of methods have been invented in the past few years to expand the number of bits for CiM. Word-line (WL) modulation, bit-serial operation, and capacitor ladder charge sharing are the three approaches adopted by numerous CiM papers.

A. Multibit Data WL Modulation

In an SRAM-based CiM scheme, because each SRAM cell can only store one digital bit, multibit weights are typically stored in a group of SRAM cells. Then, how to efficiently convert the multibit weight into an analog signal is the key problem faced by circuit designers. One way to accomplish this objective is to apply a different base to each digit by adjusting the associated WL signals of the vertically stored multibit weights. Also, there are two commonly used methods to manipulate the WL signals.

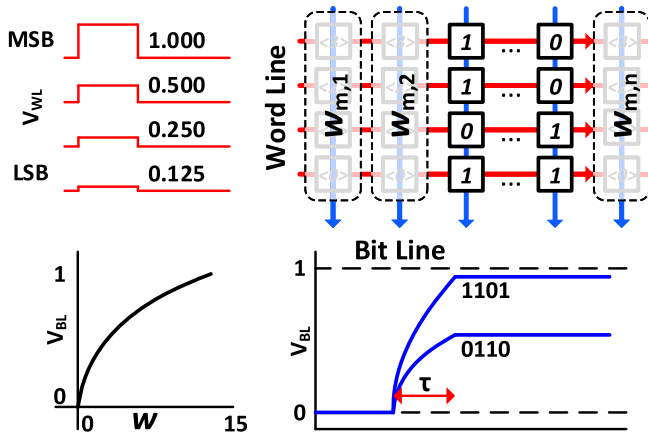


Fig. 4. Multibit weight realization with SRAM WL signal AM and its linearity performance.

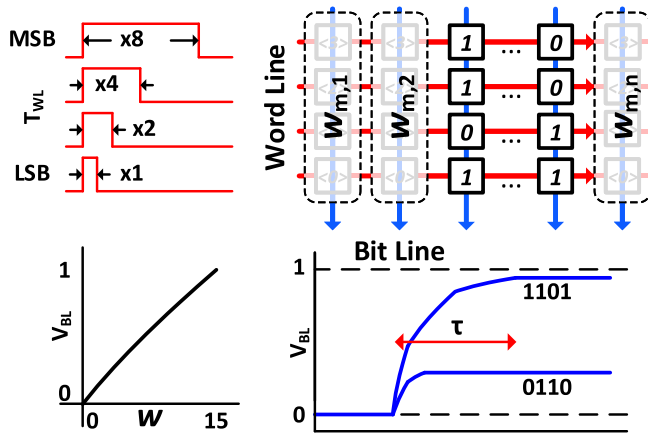


Fig. 5. Multibit weight realization with SRAM WL signal PWM and its linearity performance.

Fig. 4 shows the first solution, WL amplitude modulation (AM) [20], [21], [22]. Unlike traditional SRAM arrays where the WL signal has only two states: logic high (connected to V_{DD}) and logic low (connected to GND), the WL signals in this scheme are assigned a range of different voltage levels between V_{DD} and GND. During the computation phase, all of the WLs associated with one multibit weight are activated simultaneously. The WL associated with the most significant bit (MSB) is attached to V_{DD} , and MSB-1 is assigned a level of $V_{DD}/2$; then, the next bit is at $V_{DD}/4$ and so on. After interacting with the content stored in each SRAM cell, the signal on the attached bitline (BL) then becomes an analog voltage whose value corresponds to the binarized multibit data stored in those vertically arranged SRAM cells. Since all the cells in one weight are activated at the same time, this approach normally has a better throughput performance. However, the issue with this approach is that this WL-voltage-to-BL-voltage conversion does not adhere to a linear relationship, as shown by the transfer curve (TC) in Fig. 4. Therefore, its linearity is the major concern.

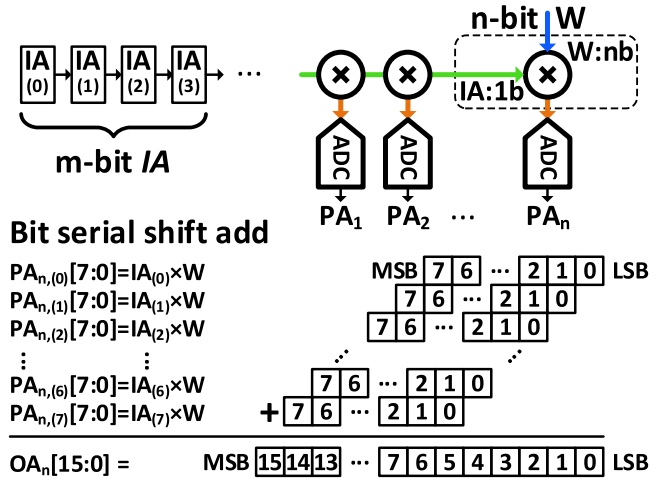


Fig. 6. Implementation of a bit-serial-based multibit multiplier.

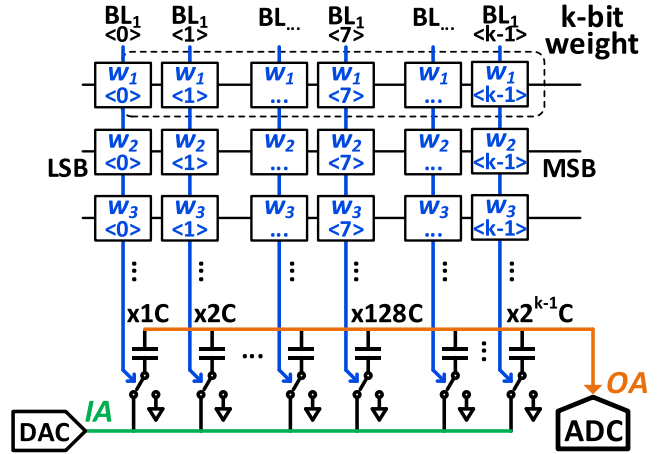


Fig. 7. Basic idea of a binary weighted capacitor ladder-based multibit multiplier for CiM.

This linearity issue can be greatly alleviated in the time domain by adopting the WL pulsewidth modulation (PWM) given in Fig. 5. Compared with the voltage AM, the linearity of the conversion between the time interval of the WL pulse and the BL voltage is significantly improved [23], [24], [25], [26]. However, it is obvious that as precision increases, the pulsewidth of the MSB grows exponentially, resulting in an unacceptable latency for high-precision data format.

B. Bit-Serial-Based Multibit Multiplication

Bit-serial operation is an approach adopted by many papers [27], [28], [29], [30], [31], [32], [33], [34] to support multibit IAs and realize multibit multiplication. In this scheme, a 1-bit-by- n -bit multiplier is utilized to process 1 bit at a time. The multiplier is normally an analog butterfly switch that swaps the polarity of the analog n -bit weight based on the 1-bit input. As presented in Fig. 6, shift and add operations are performed in the digital domain to finish the MAC operation and ensure good linearity. Since it requires multiple cycles to complete one result, it has a negative impact on the overall throughput.

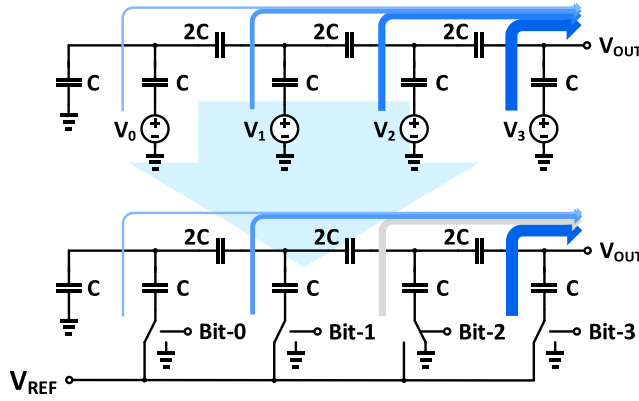


Fig. 8. Principles of the proposed C-2C capacitor ladder-based analog multibit multiplier.

C. Capacitor Ladder-Based Multibit Multiplication

The binary-weighted capacitor ladder [35], [36] is a promising approach to realize the multibit capability needed for ubiquitous edge inference applications spanning a range of accuracy requirements. As shown in Fig. 7, a k -bit ladder is a string of capacitors that consists of k -weighted capacitors in a binary weighted ratio of $1:2:4:\dots:2^{k-1}$ along the ladder. Digital words are stored in k -bit memory cells, with each cell's content controlling the corresponding branch in the ladder. The output of the ladder is then the product of the digital word applied to the ladder and the input voltage. Although this solution provides multibit capability, it is difficult to scale up, as the capacitance grows exponentially with the word length. Thus, its throughput and area efficiency degrade significantly as the precision goes up.

III. C-2C CAPACITOR LADDER-BASED LINEAR ANALOG MULTIBIT IN-MEMORY MAC UNIT

As discussed in Section II, a tradeoff between linearity, throughput, and efficiency exists in the ACiM design. To address this tradeoff, we present an area- and energy-efficient C-2C capacitor ladder scheme that is scalable for multibit CiMs in this article. The basic operating principle of this C-2C ladder-based multibit multiplication is explained in Fig. 8. Similar to the binary weight capacitor ladder, the C-2C ladder consists of a series of cascaded capacitor branches. The difference though is that instead of having binary weighted capacitors along the ladder, each branch on the C-2C ladder only contains one capacitor with one-unit capacitance C , and a serial capacitor with $2C$ is inserted between two adjacent branches (detailed in Fig. 8) [37], [38], [39]. Ratioed by the serial capacitors, the contribution of each branch is binary weighted along the ladder and superimposed at the output node of the ladder. Thus, the voltage at the output node, V_{OUT} , reflects the ON-OFF status of those switches controlled by the digital bits with a shared scaling factor of V_{REF} , as explained in the following equation:

$$V_{OUT} = V_{REF} \sum_{i=0}^3 \text{Bit}(i) \times 2^{i-4}. \quad (1)$$

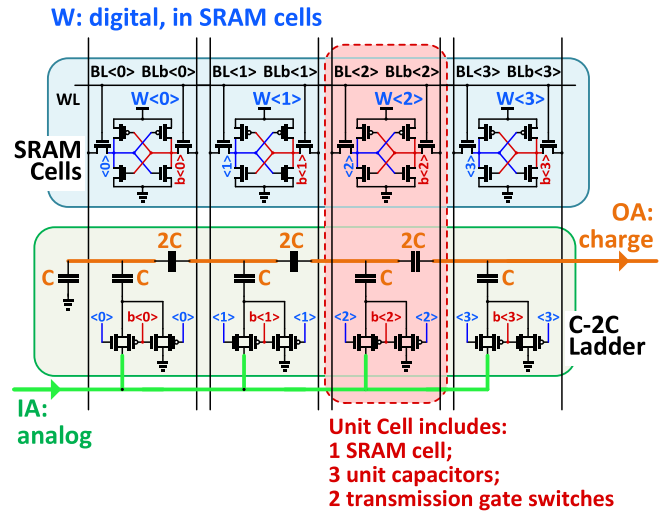


Fig. 9. Illustration of combining standard SRAM cells with the proposed C-2C ladder-based multibit multiplier.

The number of branches on the ladder corresponds to the desired word length (precision). Also, the detailed circuit implementation of our proposed structure is available in Fig. 9. As shown in Fig. 9, the upper part is a group of standard SRAM cells, while the lower part is the C-2C ladder. Transmission-gate-based switches on the ladder are controlled by digital bits stored in their paired SRAM cells and connected to either an analog input signal or to the ground depending on the stored value. When performing the MAC operation, the input signal (IA) is applied to the reference node (green wire) of the ladder, while the multibit weight (W) is stored in SRAM cells. Then, the product of those two is available at the output node (orange wire). As shown in the figure, one C-2C branch represents one digital bit. Compared with the previously mentioned binary weighted capacitor ladder, the hardware on each branch in the C-2C ladder is limited to one switch and only three unit capacitors. Thus, it is easy to scale to any number of bits required by the system. For instance, Fig. 9 contains four branches to represent a 4-bit value; then, eight branches would be eight bits or $(3 \times (8 - 1)) - 1 = 23$ unit capacitors, which is much less than the binary weighted capacitor ladder introduced in Section II.

The accumulation or summation comes next in the MAC operation. In the proposed scheme, no additional circuitry is required to carry out this step. As shown in Fig. 10(a), as multiple C-2C ladders are connected to one output node, the multiplication results are then summed and averaged through capacitive charge sharing, resulting in an analog output activation (OA), as expressed in the following equation:

$$OA = \frac{1}{m} \sum_{j=1}^m \sum_{i=0}^{k-1} IA_j \times W_{j,n<i>} \times 2^{i-k} \quad (2)$$

where m is the number of MAC units and k is weights' number of bits. In essence, each C-2C ladder is a multibit analog MAC unit. Digital-to-analog converters (DACs) and analog-to-digital converters (ADCs) are required at the input and output of the

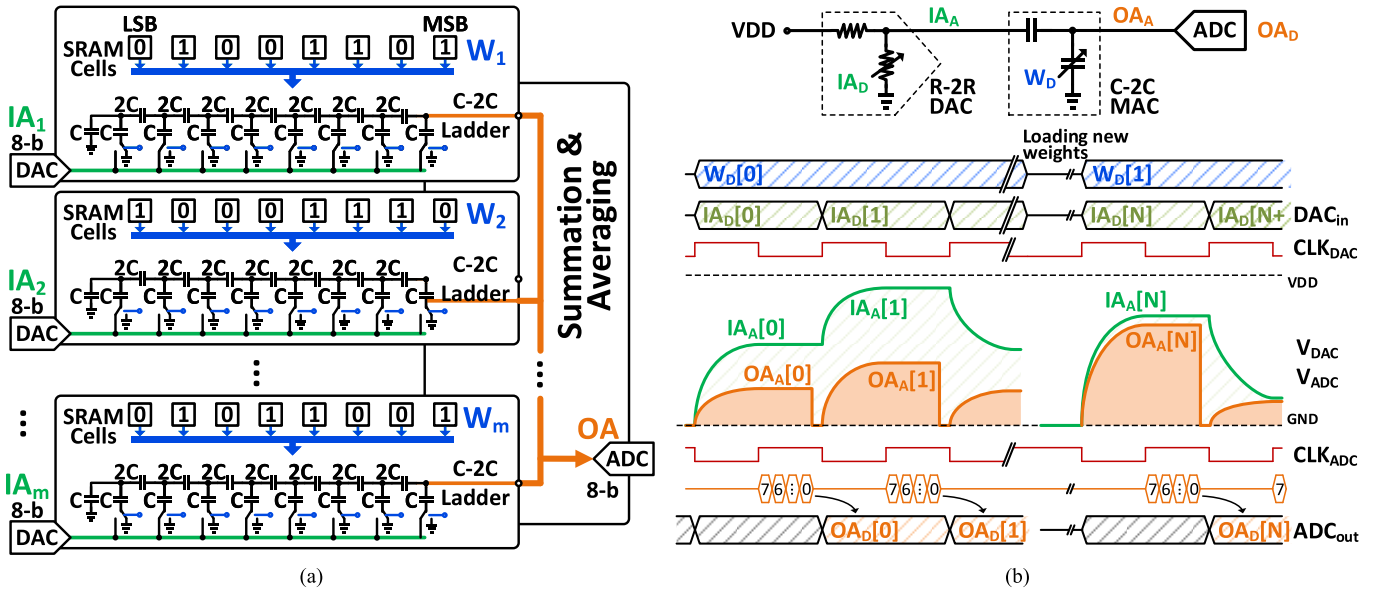


Fig. 10. Illustration of (a) proposed C-2C CiM MAC unit structure and (b) its corresponding computation timing diagram.

CiM macro to convert signals between different domains. The detailed implementation is covered in Section V.

Fig. 10(b) shows the computation flow of the proposed analog CiM macro. A two-port network with one fixed capacitor and a variable capacitor controlled by digital weights (W_D) represents the proposed C-2C in-memory MAC unit. A 1-to-2 ratioed capacitor ladder (R-2R)-based resistive DAC is used in our design to convert digital I_A (I_{A_i}) to analog I_A (I_{A_i}). The C-2C ladder divides I_{A_i} and generates O_{A_i} , which is further converted to digital O_{A_i} (O_{D_i}) by a successive-approximation ADC (SAR-ADC). As shown in the timing diagram, the DAC begins to operate at the rising edge of the DAC clock and settles before the falling edge. The C-2C ladder-based computation happens concurrently with the DAC because it is essentially an RC charging process, and once the RC network is settled, the computation is complete. The SAR-ADC then starts working on the rising edge of the ADC clock and completes all conversions before the falling edge. The clock signals of the DAC and ADC are complementary. Thus, the total computation latency, or the access time, of this analog CiM macro is only one clock cycle or about 5 ns under a 200-MHz clock.

IV. ANALOG CHARGE DOMAIN COMPUTING IMPAIRMENTS AND RESTRICTIONS

With the MAC operation performed in the analog domain, the impact of analog impairment is inevitable and could lead to significant performance degradation. In order to minimize the loss, three substantial impairment factors were considered and analyzed during the C-2C ladder circuit development: 1) parasitics on the C-2C ladder; 2) device random mismatch; and 3) noise on switched capacitors.

A. Parasitic Capacitance

If possible, a perfect 1:2 ratio is desired on the C-2C capacitor ladder. However, parasitic capacitance on the ladder

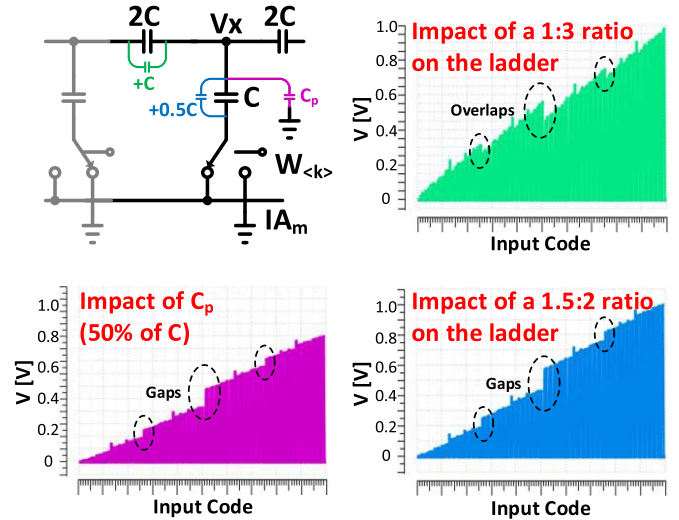


Fig. 11. Non-ideal 1:2 ratio and parasitic capacitance on the proposed C-2C capacitor ladder can cause gaps, overlaps, and dynamic range scaling on the TC.

can affect the ratio, which can lead to nonlinearity along the TC. Also, in a dense SRAM array, neither extra electromagnetic isolation nor additional capacitance calibration methods can be deployed to reduce the impact of parasitics. As shown in Fig. 11, nonideal capacitance ratios can cause gaps or overlaps on the TC, while the impact of the parasitic capacitance C_p to the ground is similar to an under ratio with a scaling factor. However, as a systematic error, some design margins can be reserved during the circuit development for error compensation. Since parasitic capacitance is inevitable, the most practical approach is to increase the size of the serial capacitor to compensate for the parasitic impact. It needs to be noted that the overlap on the TC can be mitigated by digital pre-distortion, while gaps may not.

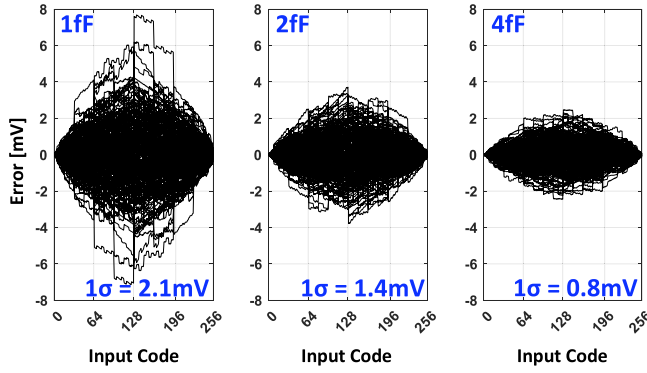


Fig. 12. C-2C ladder nonlinearity levels with different capacitor sizes due to mismatch analyzed by MC simulations.

B. Capacitor Mismatch

Device mismatch plays a critical role when using C-2C ladders to accurately convert digital weights to analog signals. While it is theoretically possible to calibrate post-fabrication capacitor mismatches, tuning each individual capacitor in the CiM array is not feasible. To maintain the accuracy, some constraints need to be placed on the minimum size of the unit capacitor, whose deviation (σ) should be less than least significant bit (LSB)/3 (e.g., <1.3 mV in an 8-bit system under 1-V dynamic range). The effect of mismatch is analyzed and quantified by Monte Carlo (MC) simulations. The deviation of a 2-fF capacitor is 1.4 mV, which stands at the margin of the LSB/3 limit, as shown in Fig. 12.

C. $k_B T/C$ Noise

The minimum size of the unit capacitor is also regulated by noise, a true random error. Although the capacitor itself does not generate thermal noise, the combination of capacitor and its series charging resistor (R) has what is called $k_B T/C$ noise, where k_B is Boltzmann's constant and T is the ambient temperature. In a sampling-based quantizer system, the capacitor's $k_B T/C$ noise needs to be lower than the quantization noise ($v_n^2 < e_q^2$) to minimize its impact. For a given precision in the system, the required minimum capacitance can be calculated by the derived equation given in the following and found in Table I

$$\left(\overline{v_n^2} = \frac{4k_B T R}{4RC_{\min}} = \frac{k_B T}{C_{\min}} \right) < \left(\overline{e_q^2} = \frac{\Delta^2}{12} \right) \quad (3)$$

then

$$\frac{k_B T}{C_{\min}} < \frac{\Delta^2}{12} \quad (4)$$

where Δ is the LSB's corresponding analog voltage step. According to Table I, a minimum 3.3-fF unit capacitor is required in an 8-bit system to have comparable impacts from both sampling noise and quantization noise. However, different from mismatch, noise's impact is applied to the output during the digitization, where multiple C-2C ladders are connected to one output node and form a much larger overall capacitance at the output. Therefore, only the overall capacitance from the

TABLE I
REQUIRED MINIMUM UNIT CAPACITANCE AT 300 K

Number of Bits	Maximum Noise in 1-V Dynamic Range	Minimum Unit Capacitance
4	18.0 mV	12.7 aF
6	4.5 mV	203.6 aF
8	1.1 mV	3.3 fF
10	281.9 μ V	52.1 fF
12	70.5 μ V	833.9 fF
14	17.6 μ V	13.3 pF
16	4.4 μ V	213.5 pF

C-2C ladder array needs to satisfy the minimum capacitance requirement.

V. PROPOSED C-2C LADDER CiM MACRO IMPLEMENTATION

The chip is split into two identical analog CiM cores, the west core and the east core, as shown in Fig. 13. Each core contains an analog CiM macro, which has been detailed on the right of the figure. Essentially, it is a 64-row $\times$ 16-column array of the proposed 8-bit C-2C MAC unit, along with 64 8-bit DACs and 16 8-bit ADCs. Each row takes one analog IA from an 8-bit DAC and fan-out to 16 MAC units in those 16 columns. Also, each of the 16 columns collects the charge summation from 64 MACs and digitizes the result with an 8-bit SAR-ADC. The rest of the chip is occupied by a lightweight digital controller, shared memory (SMEM) for buffering off-chip data, I/O banks (serial peripheral interface (SPI) and field-programmable gate array (FPGA) interfaces), clock generation/distribution, voltage regulators, and other testing modules.

A. Weight Local Multiplexing

As illustrated in Section III, the number of components in a C-2C ladder is compatible with 6T SRAM cells. Moreover, since the capacitors are formed with back-end-of-line (BEOL) metal-oxide-metal (MoM) structures [40], [41], they can be placed above the SRAM array to minimize silicon area impact, as shown in Fig. 14(a). Components circled by red boxes are active devices on silicon, while those circled in blue are implemented with BEOL structures.

To reduce the dead time associated with loading new weights, a sub-bank multiplexing scheme is adopted that both improves the weight stationary rate and reduces the required memory write bandwidth. As presented in Fig. 14(b) and (c), eight parallelly arranged 8-bit digital words in SRAM cells are assigned to one 8-bit C-2C ladder by using the multiplexing structure. A pair of compute-WL (CWL) and compute-BL (CBL) are added to the memory cell to locally select one of the eight sub-banks in the array every cycle. To decouple the SRAM's write/read operation from the CiM MAC operation, three additional transistors are attached to a 6T SRAM cell,

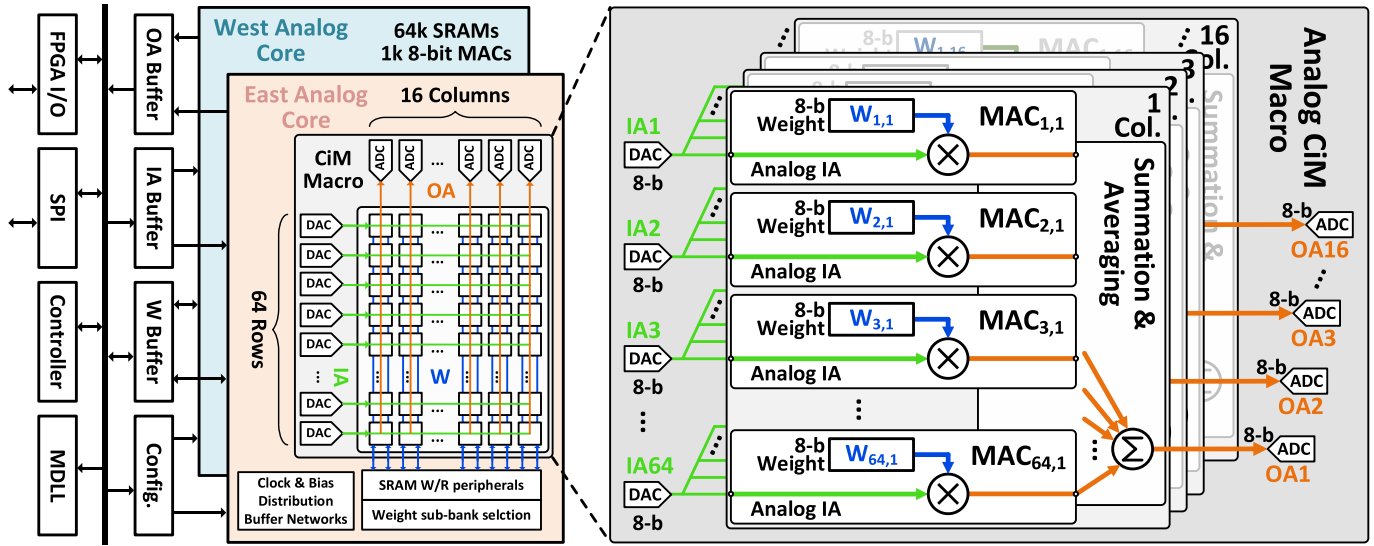


Fig. 13. Chip-level architecture diagram of the proposed analog CiM and the detailed analog CiM macrostructure with peripherals.

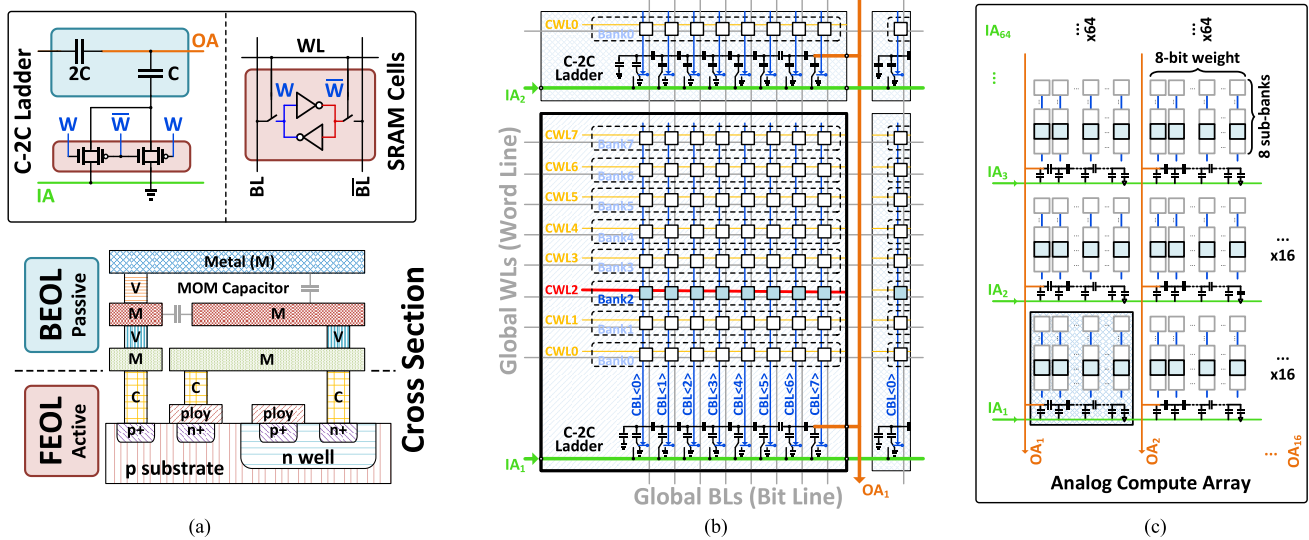


Fig. 14. (a) Connection between C-2C ladder and SRAM cell, as well as the stacking layout consideration. (b) Proposed local weight sub-bank multiplexing structure. (c) CiM array in each analog core.

as shown in Fig. 15, which allows reuse of the DAC input. It also provides additional isolation and protects the content stored in the SRAM cell's latch from unwanted noise picked up by the CBL in the array. The reliability of the newly proposed 9T SRAM cell has been verified by performing MC static-noise-margin (SNM) tests over different process corners. According to the simulation results, the SNM degradation due to asymmetric loading is less than 2% across all the corners. With the proposed sub-bank multiplexing, one C-2C ladder is shared by multiple SRAM cells, which lifts the size constraints applied to the unit capacitor in the ladder and improves overall memory density. As each analog macro contains 1k 8-bit C-2C ladder-based MAC units, the overall SRAM memory size in the CiM array is 64 kb.

B. CiM SRAM Array

The CiM array with the proposed 9T SRAM cell is converted from an autogenerated 6T SRAM array using Intel's

in-house SRAM generation tool for the Intel 22FFL process. Fig. 16 shows one example of the conversion process where the WL and BL logic as well as SRAM IO logic remain intact in the autogenerated SRAM array, whereas the 6T cell array has been replaced with 9T cell clusters. The 9T cell has exactly $2\times$ the width of the 6T cell for pitch matching purposes, resulting in the removal of half of the WLs. Each cluster consists of eight SRAM cells, one C-2C unit, and control logic, representing a 1-bit weight with eight sub-banks. Eight clusters are then arrayed vertically in the layout to form the 8-bit weight structure. This CiM array conversion process is carried out manually in this prototype. However, an automation flow has been envisioned and researched for future development.

C. Analog CiM Peripherals

Although data converters are normally considered as peripheral circuitry in analog CiM systems, they have a strong impact

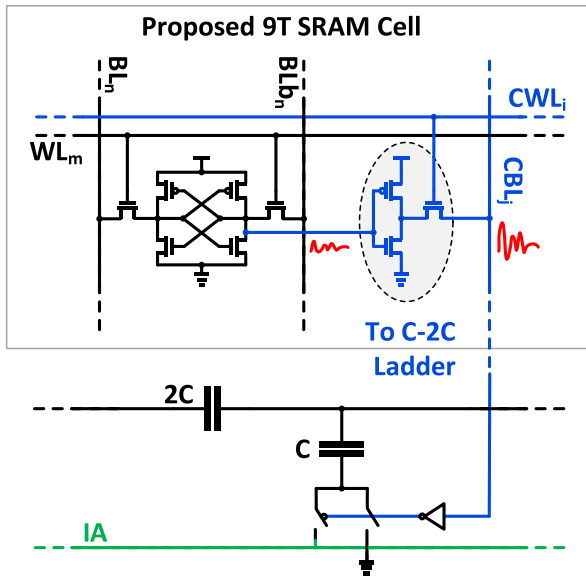


Fig. 15. Proposed 9T SRAM cell with one C-2C ladder branch for multibit CiM.

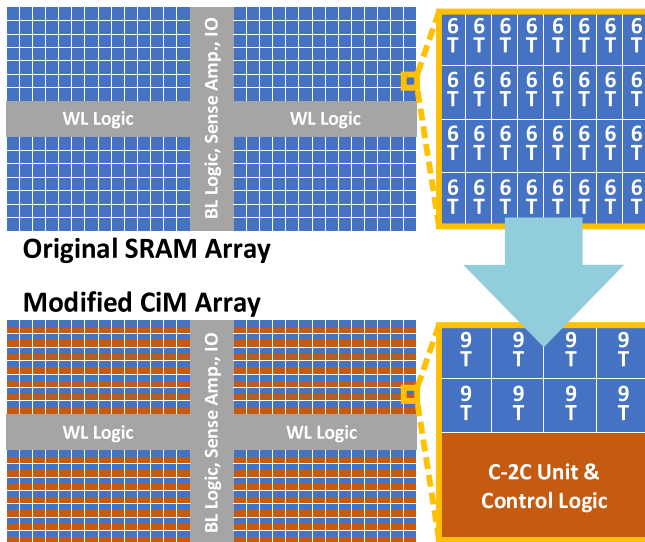


Fig. 16. Illustration layout of the SRAM array in Intel 22FFL process and the manually converted 9T SRAM-based CiM array with peripheral reuse.

on the overall performance of the CiM implementation in terms of conversion linearity, computation accuracy, power, and area efficiency. Low-cost DAC and ADC structures are adopted in this design.

In this work, C-2C capacitor ladders acting as a DAC have been used for digital weight conversion and multiplication, while another DAC is needed to provide an analog IA signal to the reference node of the C-2C ladder. An intuitive thought is to use another C-2C ladder for the IA conversion. However, without any driving capability, the C-2C ladder cannot fan-out to multiple MAC units, as shown in Fig. 17(a). The equivalent capacitance seen in the output node of the C-2C ladder is equal to twice of the unit capacitance, or $2C$, on the ladder, while as shown in Fig. 17(b), the equivalent capacitance seen

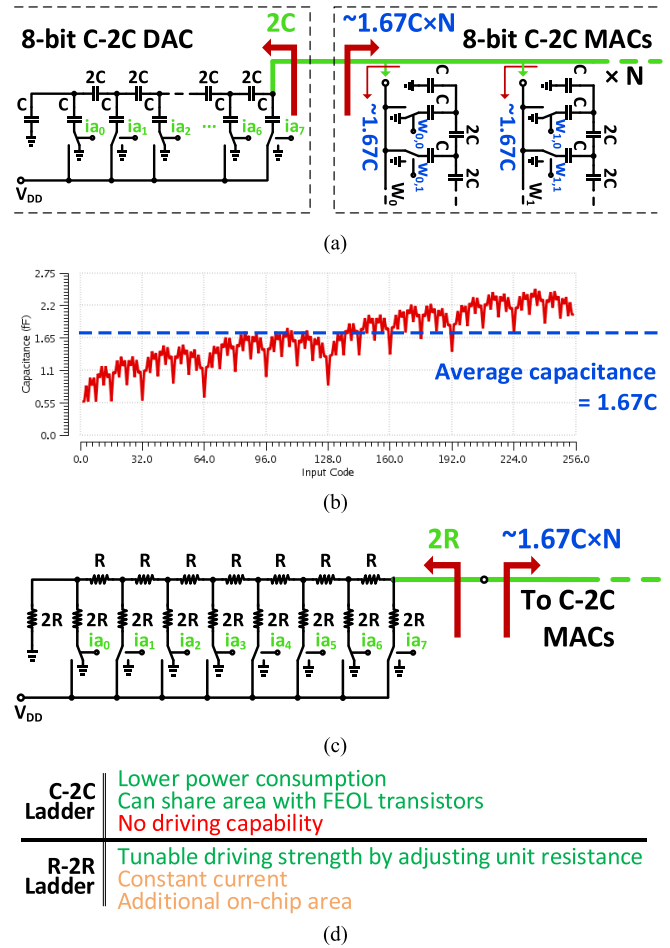


Fig. 17. Digital-to-analog conversion schemes. (a) C-2C ladder-based conversion and its practical problems. (b) Code-dependent equivalent capacitance seen at the reference node of a C-2C ladder. (c) Proposed R-2R ladder-based solution with sufficiency-driven capability. (d) Comparison between C-2C ladder and R-2R ladder.

in the reference node of the C-2C ladder is code dependent with an average value of $1.45C$. A direct connection can only result in a charge redistribution and an impractical dynamic range. Therefore, an 8-bit R-2R resistor ladder [Fig. 17(c)] with sufficient current driving strength is then selected in this work to drive multiple C-2C ladders in the analog MAC array. The benefits and drawbacks of both the C-2C ladder and the R-2R ladder have been listed in Fig. 17(d).

After the analog MAC operation, a low-power, 8-bit asynchronous SAR-ADC is utilized to efficiently convert the analog voltage stored on the charge summation node to digital bits. As presented in Fig. 18, the ADC in this implementation was developed based on the work published in [37], which replaces the binary capacitive DAC inside the conventional ADC with a C-2C ladder DAC and is controlled by the SAR logic to perform a binary search. The built-in asynchronous SAR clock reduces power consumption while achieving an acceptable conversion rate. The internal clock starts at the rising edge of the external clock and generates eight pulses within one external clock period. In the first internal cycle, the MSB register in the SAR logic is set to "1" and all the

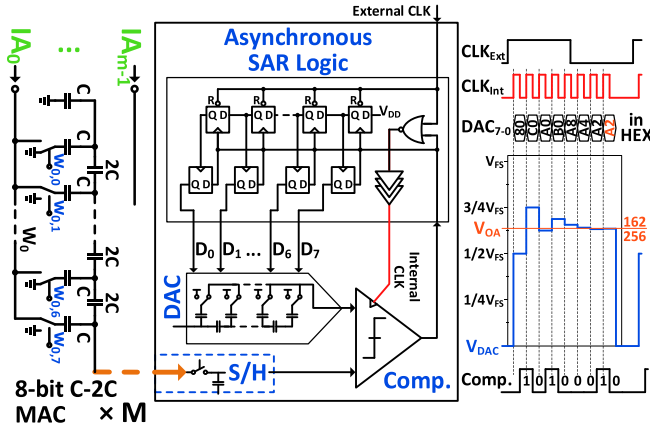


Fig. 18. Analog output digitization with a directly connected 8-bit asynchronous SAR ADC and its corresponding SAR operation.

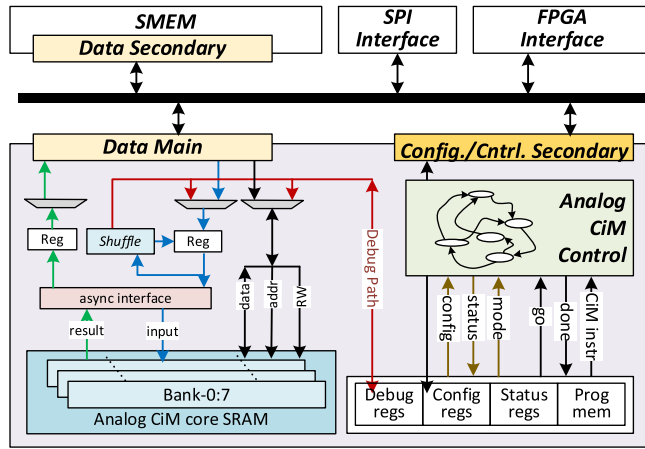


Fig. 19. On-chip digital controller with a custom, lightweight 10-bit instruction set for data movement, arithmetic operations, and configuration.

remaining bits are set to “0” or “80” in hexadecimal (HEX) format for an 8-bit system. The DAC takes this code and generates the corresponding analog voltage, which is then compared with the sampled input. If it is lower than the input, then the bit stays at “1”; otherwise, the SAR logic resets this bit to “0.” Then, MSB-1 is set to “1” and the same test is done, continuing this binary search until all the bits have been tested. As shown in the figure, the output of the DAC is gradually approaching the sampled voltage. The sample-and-hold (S/H) in the design reuses the combined C-2C ladders in the CiM array to further reduce the overall area overhead. Since 64 C-2C ladders are connected together to form the S/H, a total 128-fF input capacitance is adequate to meet the noise requirement for an 8-bit ADC according to the kT/C equation.

D. System Controller

The digital core uses a custom, lightweight 10-bit instruction set to perform data movement, arithmetic operations, and configuration, as shown in Fig. 19. The instructions support several modes of operation. The MVM of a fully connected (FC) layer is executed by loading weights into the SRAM banks and fetching input data into a local buffer. Successively

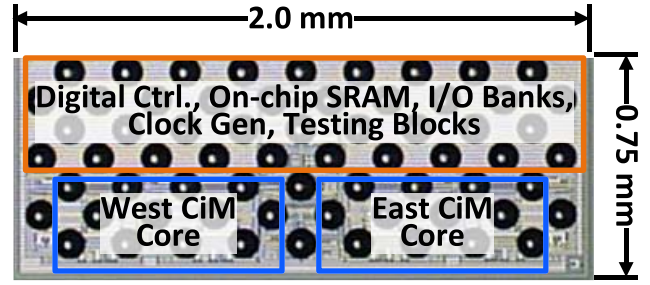


Fig. 20. Micrograph of the proposed analog CiM test chip.

switching between banks allows for input reuse. Similarly, 1-D/2-D convolutional layers are executed by storing input feature maps in the SRAM banks and fetching filter coefficients into the local buffer. The filter coefficients can be shifted ± 1 or ± 2 in X/Y in tandem within the local buffer in combination with re-executing MVM. The analog CiM SRAM macros can perform an 8-bit 64×16 MVM in a single clock cycle by using a total of 64k bits of SRAM spread across eight sub-banks. The tiles are linked together via an on-chip system fabric with a high-speed serial link for off-chip communication and configuration via an external FPGA board. The latency, or the access time, of this analog CiM macro is one clock cycle or about 5 ns under a 200-MHz clock. First, the DAC begins to run at the rising edge of the clock and settles before the falling edge. The C-2C ladder-based computation happens concurrently with the DAC because it is essentially an RC charging process, and once the RC network is settled, the computation is complete. The SAR-ADC then starts working on the falling edge of the clock and completes all conversions before the next rising edge.

VI. MEASUREMENT RESULTS

The prototype was fabricated using the Intel 22FFL technology. Fig. 20 shows the die photograph of the chip. The overall chip area is 1.5 mm^2 , while each of the analog CiM core only occupies an area of 0.25 mm^2 . The chip works under a 0.7–1.1-V power supply, at 145–240 MHz. An average power of 54.8 mW is consumed by the two CiM cores at 220 MHz and 1.0 V. Also, it supports the INT-8 data format for IAs, weights, and outputs. The default chip specifications have been detailed in Table II. Fig. 21 shows the block diagram of the chip test setup for measurement and demonstration. Two sets of power supply and clock signals are fed into the chip separately, in order to better evaluate and calculate the energy efficiency and throughput of the analog CiM macro. The chip can be programmed either through an FPGA evaluation board or via a standard SPI interface.

Each analog CiM core includes 64 DACs, 16 ADCs, 1k C-2C MAC units, 64-kb SRAM cells, and supporting circuits, including clock distribution buffers, I/O buffers, and decoupling capacitors. A detailed area and power breakdown are given in Fig. 22(a). Unlike bit-wise multiplication, the proposed C-2C ladder analog CiM scheme resolves multibit MAC operation in only one clock cycle, which leads to

TABLE II
CHIP SUMMARY

Technology	Intel 16 CMOS
Chip Area	1.5 mm ²
Core Area	0.124 mm ² × 2
Digital Clock	800 MHz
Analog Clock	200 MHz
Bit Precision (I/W/O)	INT8/ INT8/ INT8
Number of MACs	2048
CiM Memory	128 kb
On-chip Memory	2 Mb

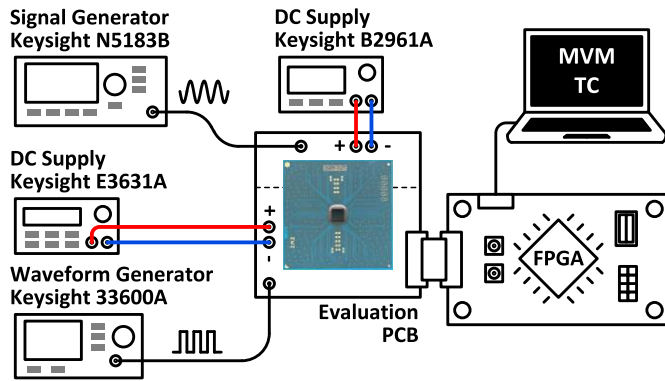
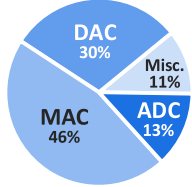
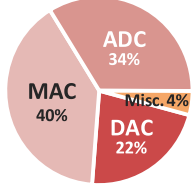


Fig. 21. Block diagram of the chip testing setup with two separated power and clock domains for accurate performance evaluation.

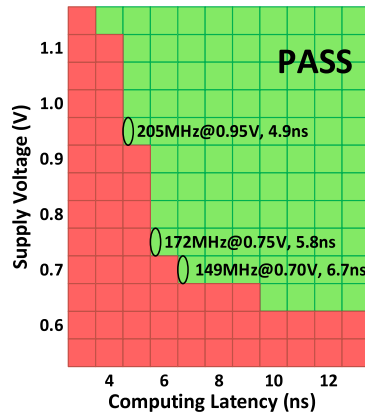
CiM Core Area Breakdown



CiM Core Power Breakdown



(a)



(b)

Fig. 22. (a) Area and power breakdown of the analog CiM macro. (b) Measured Shmoo plot with access time of the analog CiM with 8-bit IAs and weights.

exceptional power and area efficiency. As a mixed-signal design, its performance highly relies on the voltage supply. The maximum clock rate of the analog CiM core is labeled on the measured Shmoo plot in Fig. 22(b). A peak energy efficiency of 32.2 TOPS/W is achieved with a 0.7-V power

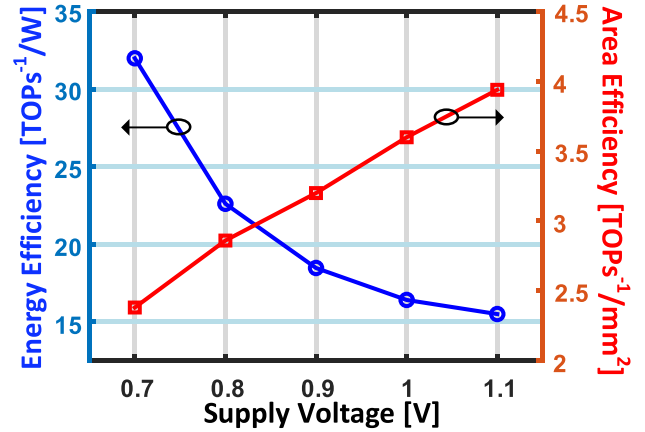


Fig. 23. Measured energy efficiency and area efficiency versus supply voltage with no sparsity employed.

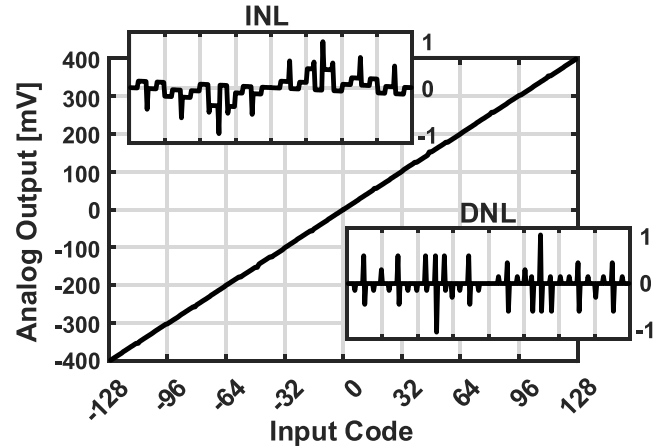


Fig. 24. Measured DAC standalone TC, INL, and DNL plots.

supply and no sparsity, while the highest 4.0-TOPS/mm² area efficiency is reached under 1.1 V, as presented in Fig. 23.

Computation accuracy is one of the key indicators for analog CiM designs to evaluate their performance. The major error comes from nonlinearity. Since both the R-2R ladder-based DAC and the C-2C ladder-based in-memory MAC unit implementations preclude the need for any active component for the data conversion and the arithmetic operations, this leads to great linearity performance. The DAC stand-alone linearity TC was captured at an analog observation point reserved before the ADC. Its corresponding integral nonlinearity (INL) and differential nonlinearity (DNL) were calculated based on the measured TC, as shown in Fig. 24. The linearity of the SAR-ADC cannot be measured directly, as there is no analog input port reserved in this work. However, its performance can be inferred by the measured overall digital-to-digital conversion results. Fig. 25(a) gives two TCs: 1) fix the weight to the minimum code (−127) and sweep the IA code and 2) fix the IA to the maximum code (+127) and sweep the weight code. The calculated DNL and INL curves are available in Fig. 25(b). They are both lower than three LSBs, which leads

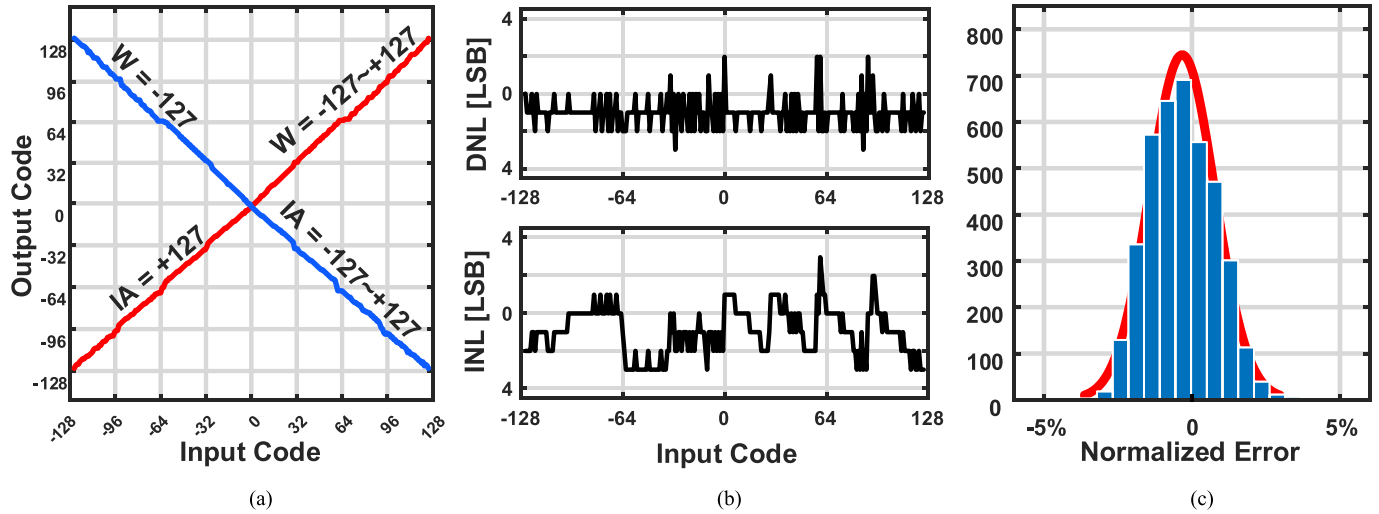


Fig. 25. (a) Measured TCs of the test chip when fixing the weight code to -127 and sweeping the IA code and fixing the IA code to 127 and sweeping the weight code. (b) Corresponding calculated DNL and INL based on the measured TC. (c) Error distribution of a 10k-point random MVM test result, showing a 0.89% computation error.

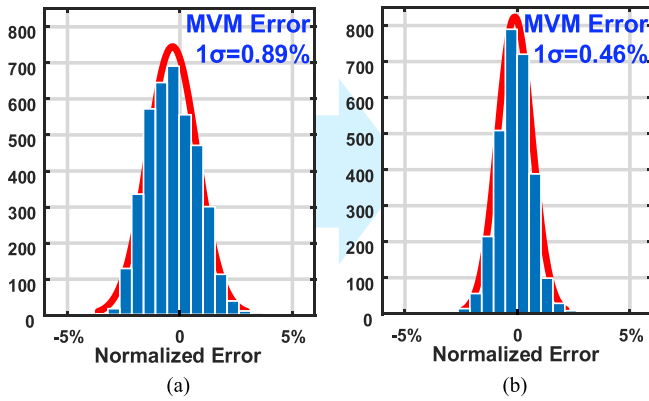


Fig. 26. (a) Measured uncompensated error distribution of a 10k-point random MVM test. (b) Improved distribution with input pre-distortion.

to great error distribution performance during a 10k-point random MVM test. A 1σ error of 0.89% is achieved, as shown in the distribution plot in Fig. 25(c). It is the lowest so far in all the published analog computing implementations. As a systematic error was observed along the TC, pre-distortion can be applied to the input code. Once the nonlinearity is compensated, the error distribution becomes less than 0.5%, as shown in Fig. 26. However, the price is a reduced overall dynamic range.

To validate the inference capability of the analog CiM macro, a four-layer multilayer perceptron (MLP) and a compressed LeNet-5 CNN network [42], [43] were trained to perform the MNIST image classification. The integrated controller was configured to support both the FC layer and the convolutional (Conv.) layer. For the MLP, the reference software accuracy is 98.26% on the MNIST dataset, and our hardware achieves 98.14%. Also, for the compressed LeNet-5, the reference software accuracy is 98.8%, and our hardware achieves 96.85%.

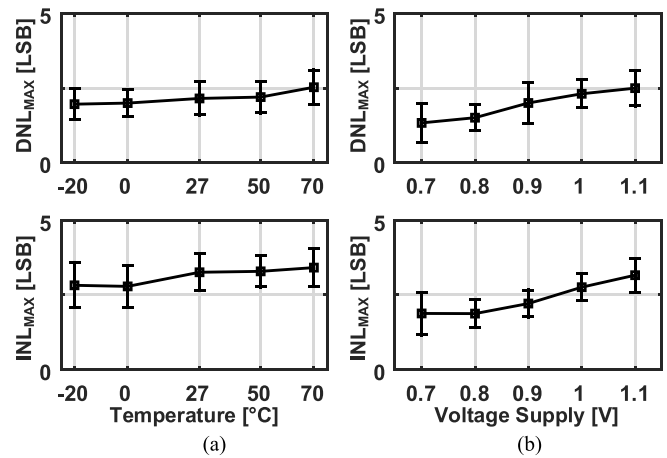


Fig. 27. (a) INL and DNL plots based on measured TCs from five test chips under different temperatures. (b) INL and DNL plots based on measured TCs from five test chips under different supply voltages.

Usually, analog and mixed-signal circuits suffer from process–voltage–temperature (PVT) variations. However, passive components, such as capacitors, are more resilient to PVT variations than active components. Due to the proposed passive C-2C ladder structure, the measured results, shown in Fig. 27, are very consistent and promising over PVT, which indicates a good PVT tolerance. We measured 160 TCs and calculated the corresponding DNL_{MAX} and INL_{MAX} numbers from all the ADCs on five chips (each chip contains 32 ADCs) under a wide temperature and voltage range as shown in Fig. 27.

VII. CONCLUSION

An SRAM-based analog charge domain CiM macro is presented in this article with the Intel 22FFL process. Multibit precision is realized by a unique C-2C capacitor ladder-based computing scheme. The performance of the CiM varies under different supply voltages and achieves a peak energy efficiency

TABLE III
PERFORMANCE SUMMARY COMPARISON

Parameter	This Work		JSSC' 21 [44]	ISSCC' 21 [36]	JSSC' 22 [45]	ISSCC' 22 [46]	ISSCC' 22 [47]	JSSC' 22 [48]	ISSCC' 22 [49]
Topology	22nm FFL ⁽¹⁾		65 nm	65 nm	16 nm	5 nm	28 nm	14 nm	28 nm
Core Area [mm ²]	0.25		0.17	0.57	ca.16	0.0133	0.94	0.63	-
Memory Type	SRAM		SRAM	eDRAM	SRAM	SRAM	SRAM	PCM	SRAM
Cell Structure	9T		6T	1T1C	8T	12T	6T	8T4R	6T
CiM Memory	128kb		64kb	16kb	4.5Mb	64kb	96kb	64kb	1Mb
Computing Mechanism	C-2C Charge		Charge	Charge	Charge	Digital	Digital Booth	Current	Time
Input Bitwidth	8		4	8	8	4	8	8	8
Weight Bitwidth	8		8	8	8	4	8	4 ⁽⁷⁾	8
Output Bitwidth	8		7	8	8	14	18	8	22
Computation Error	0.89%, 0.46% ⁽³⁾		1.47%	4.17% ⁽⁶⁾	0.94% ⁽⁶⁾	0%	0%	1.94%	-
Effective Memory Density [bit/mm ²]	512k		376.5k	28.1k	281.3k	4.8M	102.1k	101.6k	-
Voltage Supply [V]	0.7	1.1	1.2	1-1.2	0.8	0.5-0.9	0.6-1.0	0.8	0.65-0.9
Clk Freq. [MHz]	145	240	70	50-200	200	360-1440	50-220	1000	150 ⁽⁸⁾
Access Time [ns]	6.9	4.2	14.3	10-40	5	2.8	4.5	128	6.6-7.8
Throughput ⁽²⁾ [TOPs ⁻¹] (I:W)	0.6 (8b:8b)	1.0 (8b:8b)	0.6 (4b:2b)	0.005 (8b:8b)	3.0 (8b:8b)	2.95 (4b:4b)	1.35 (8b:8b)	1.0 (8b:4b)	1.24 (8b:8b)
Energy Efficiency ⁽²⁾ [TOPs ⁻¹ /W] (I:W)	32.2 (8b:8b)	15.5 (8b:8b)	49.4 (4b:1b)	4.76 (8b:8b)	30 (8b:8b)	205.0 (4b:4b)	57.8 (8b:8b)	10.5 (8b:4b)	27.75 ⁽⁹⁾ (8b:8b)
Normalized ⁽⁴⁾ TOPs ⁻¹ /W	2061	992	449	692	1509	1069	4440	239	2132
Area Efficiency ⁽²⁾ [TOPS/mm ²] (I:W)	2.4 (8b:8b)	4.0 (8b:8b)	3.4 (4b:2b)	0.008 (8b:8b)	2.67 (4b:4b)	221.2 (4b:4b)	1.4 (8b:8b)	1.59 (8b:4b)	-
Normalized ⁽⁵⁾ TOPs ⁻¹ /mm ²	154	256	171	3.2	25	287	135	24	-
Sparsity (W:IA)	0%:50%		-	-	-	75%:75%	0%:0%	-	90%:90%
Supporting Formats	INT-8		INT-8	INT-8	INT-8	INT-4	INT-8, FP	INT-8	INT-8
Neural Network Models	MLP LeNet-5		LeNet-5 ResNet-20	VGG-like	VGG ResNet-50	-	ResNet-50 EfficientNet	MLP ResNet-9	ResNet-20
Classification Datasets	MNIST		MNIST CIFAR-10	CIFAR-10	CIFAR-10 ImageNet	-	ImageNet	MNIST CIFAR-10	CIFAR-10 CIFAR-100
Inference Accuracy [%]	98.14% 96.85%		98.8% 89.0%	83.27%	91.51% 73.33%	-	75.87%, 76.23%	98.3%, 85.6%	92.08%, 67.81%

- Intel 22FFL stands for "Intel 22nm FinFET Low Power" process and has been upgraded to "Intel 16" for Intel IFS customers.
- Reported number with default input precision and weight precision.
- Equivalent parts total power.
- Normalized to 1-bit input and weight. Scaled to 22 nm, assuming 30% energy efficiency improvement per technology node. Equation: $\text{TOPS/W} \cdot \text{bitwidth(I)} \cdot \text{bitwidth(W)} \cdot (1+30\%)^{\log_2((\text{node}^2/22\text{nm}^2))}$

- Normalized to 1-bit input and weight. Scaled to 22 nm, assuming 80% area efficiency improvement per technology node. Equation: $\text{TOPS/mm}^2 \cdot \text{bitwidth(I)} \cdot \text{bitwidth(W)} \cdot (1+80\%)^{\log_2((\text{node}^2/22\text{nm}^2))}$
- Estimated from measured transfer curves of DAC and ADC.
- Estimated number of bits based on the result given in the paper.
- Estimated clock frequency from reported access time.
- Performance in 90% input sparsity.

of 32.2 TOPS/W at 0.7 V and a peak area efficiency of 4.0 TOPS/mm² at 1.1 V. Several critical analog impairment factors (parasitics, mismatch, and noise) have been considered during the circuit development, which ensures a sufficiently high multibit linearity and leads to exceptional computation accuracy performance in MVM tests. The detailed performance is summarized in Table III and compared with other recently published state-of-the-art multibit analog and digital CiM implementations. Furthermore, due to the proposed passive C-2C ladder-based analog computing mechanism,

a consistent linearity and computation accuracy have been validated under different operation conditions, indicating a good PVT tolerance.

ACKNOWLEDGMENT

The authors would like to acknowledge Enrique "Max" Calderon, Erika Ramirez Lozano, Cuong Le, Brando Perez Esparza, Sandeep Chippada, Sirisha R. Kale, Jonathan Almanza Batres, Yuyun Liao, Asma Beevi Kuriparambil Thekkumpate, Xiaosen Liu, May Wu, Haifa

Qistina Hassan Nordin, Kin Wai Lee, and Divya Shree Vemparala for modeling, tape-out, packaging, and testing support.

REFERENCES

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, May 2016, pp. 770–778.
- [3] D. Amodei et al., "Deep speech 2: End-to-end speech recognition in English and Mandarin," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, vol. 48, Jun. 2016, pp. 173–182.
- [4] A. W. Senior et al., "Improved protein structure prediction using potentials from deep learning," *Nature*, vol. 577, no. 7792, pp. 706–710, 2020.
- [5] B. Schmauch et al., "A deep learning model to predict RNA-seq expression of tumours from whole slide images," *Nature Commun.*, vol. 11, no. 1, pp. 1–15, Aug. 2020.
- [6] J. Konecny et al., "Federated learning: Strategies for improving communication efficiency," in *Proc. NIPS*, 2016, pp. 1–10.
- [7] B. McMahan, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, 2017, pp. 1273–1282.
- [8] J. Konecny, H. B. McMahan, D. Ramage, and P. Richtarik, "Federated optimization: Distributed machine learning for on-device intelligence," 2016, *arXiv:1610.02527*.
- [9] S. Augenstein et al., "Generative models for effective ML on private, decentralized datasets," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019, pp. 1–26.
- [10] C. He, C. Tan, H. Tang, S. Qiu, and J. Liu, "Central server free federated learning over single-sided trust social networks," 2019, *arXiv:1910.04956*.
- [11] Y. Chen et al., "DaDianNao: A machine-learning supercomputer," in *Proc. 47th Annu. IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, Dec. 2014, pp. 609–622.
- [12] B. Moons, R. Uytterhoeven, W. Dehaene, and M. Verhelst, "Envision: A 0.26-to-10 TOPS/W subword-parallel dynamic-voltage-accuracy-frequency-scalable convolutional neural network processor in 28 nm FDSOI," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 246–247.
- [13] Y. H. Chen, T. Krishna, J. S. Emer, and V. Sze, "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 127–138, Jan. 2017.
- [14] V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proc. IEEE*, vol. 105, no. 12, pp. 2295–2329, Dec. 2017.
- [15] Y. Kim, D. Shin, J. Lee, Y. Lee, and H.-J. Yoo, "A 0.55 V 1.1 mW artificial-intelligence processor with PVT compensation for micro robots," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Jan. 2016, pp. 258–259.
- [16] H. Valavi, P. J. Ramadge, E. Nestler, and N. Verma, "A 64-tile 2.4-Mb in-memory-computing CNN accelerator employing charge-domain compute," *IEEE J. Solid-State Circuits*, vol. 54, no. 6, pp. 1789–1799, Jun. 2019.
- [17] S. Yin, Z. Jiang, J.-S. Seo, and M. Seok, "XNOR-SRAM: In-memory computing SRAM macro for binary/ternary deep neural networks," *IEEE J. Solid-State Circuits*, vol. 55, no. 6, pp. 1733–1743, Jun. 2020.
- [18] C. Yu, T. Yoo, K. T. C. Chai, T. T.-H. Kim, and B. Kim, "A 65-nm 8T SRAM compute-in-memory macro with column ADCs for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 57, no. 11, pp. 3466–3476, Nov. 2022.
- [19] H. Valavi, P. J. Ramadge, E. Nestler, and N. Verma, "A mixed-signal binarized convolutional-neural-network accelerator integrating dense weight storage and multiplication for reduced data movement," in *Proc. IEEE Symp. VLSI Circuits*, Jun. 2018, pp. 141–142.
- [20] J. Zhang, Z. Wang, and N. Verma, "In-memory computation of a machine-learning classifier in a standard 6T SRAM array," *IEEE J. Solid-State Circuits*, vol. 52, no. 4, pp. 915–924, Apr. 2017.
- [21] J.-W. Su et al., "A 28 nm 64 Kb inference-training two-way transpose multibit 6T SRAM compute-in-memory macro for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 240–242.
- [22] J. Yue et al., "A 65 nm computing-in-memory-based CNN processor with 2.9-to-35.8 TOPS/W system energy efficiency using dynamic-sparsity performance-scaling architecture and energy-efficient inter/intra-macro data reuse," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 234–236.
- [23] M. Kang, M.-S. Keel, N. R. Shanbhag, S. Eilert, and K. Curewitz, "An energy-efficient VLSI architecture for pattern recognition via deep embedding of computation in SRAM," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 8326–8330.
- [24] A. Biswas and A. P. Chandrakasan, "Conv-RAM: An energy-efficient SRAM with embedded convolution computation for low-power CNN-based machine learning applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 488–490.
- [25] M. Kang, S. K. Gonugondla, A. Patil, and N. R. Shanbhag, "A multi-functional in-memory inference processor using a standard 6T SRAM array," *IEEE J. Solid-State Circuits*, vol. 53, no. 2, pp. 642–655, Feb. 2018.
- [26] S. K. Gonugondla, M. Kang, and N. R. Shanbhag, "A Variation-tolerant in-memory machine learning classifier via on-chip training," *IEEE J. Solid-State Circuits*, vol. 53, no. 11, pp. 3163–3173, Nov. 2018.
- [27] P. Judd, J. Albericio, T. Hetherington, T. M. Aamodt, and A. Moshovos, "Stripes: Bit-serial deep neural network computing," in *Proc. 49th Annu. IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, Oct. 2016, pp. 1–12.
- [28] R. Dorrance et al., "Energy efficient BNN accelerator using CIM and a time-interleaved Hadamard digital GRNG in 22 nm CMOS," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, 2022, pp. 2–4.
- [29] H. Kim, T. Yoo, T. T.-H. Kim, and B. Kim, "Colonnade: A reconfigurable SRAM-based digital bit-serial compute-in-memory macro for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 56, no. 7, pp. 2221–2233, Jul. 2021.
- [30] H. Kim, Q. Chen, T. Yoo, T. T.-H. Kim, and B. Kim, "A 1–16 b precision reconfigurable digital in-memory computing macro featuring column-MAC architecture and bit-serial computation," in *Proc. IEEE 45th Eur. Solid State Circuits Conf. (ESSCIRC)*, Sep. 2019, pp. 345–348.
- [31] J. Wang et al., "A 28-nm compute SRAM with bit-serial logic/arithmetic operations for programmable in-memory vector computing," *IEEE J. Solid-State Circuits*, vol. 55, no. 1, pp. 76–86, Jan. 2020.
- [32] Y.-C. Chiu et al., "A 22 nm 4 Mb STT-MRAM data-encrypted near-memory computation macro with a 192GB/s read-and-decryption bandwidth and 25.1–055.1 TOPS/W 8 b MAC for AI operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 178–180.
- [33] W.-S. Khwa et al., "A 40-nm, 2 M-cell, 8 b-precision, hybrid SLC-MLC PCM computing-in-memory macro with 20.5–65.0 TOPS/W for tiny-al edge devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.
- [34] J.-M. Hung et al., "An 8-Mb DC-current-free binary-to-8 b precision ReRAM nonvolatile computing-in-memory macro using time-space-readout with 1286.4–21.6 TOPS/W for edge-AI devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.
- [35] M. E. Sinangil et al., "A 7-nm compute-in-memory SRAM macro supporting multi-bit input, weight and output and achieving 351 TOPS/W and 372.4 GOPS," *IEEE J. Solid-State Circuits*, vol. 56, no. 1, pp. 188–198, Jan. 2021.
- [36] S. Xie, C. Ni, A. Sayal, P. Jain, F. Hamzaoglu, and J. P. Kulkarni, "eDRAM-CIM: Compute-in-memory design with reconfigurable embedded-dynamic-memory array realizing adaptive data converters and charge-domain computing," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 248–250.
- [37] E. Alpmann, H. Lakdawala, L. R. Carley, and K. Soumyanath, "A 1.1 V 50 mW 2.5 GS/s 7 b time-interleaved C-2C SAR ADC in 45 nm LP digital CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2009, pp. 76–77.
- [38] H. Wang et al., "A 32.2 TOPS/W SRAM compute-in-memory macro employing a linear 8-bit C-2C ladder for charge domain computation in 22 nm for edge inference," in *Proc. IEEE Symp. VLSI Technol. Circuits*, Jun. 2022, pp. 36–37.
- [39] B. Razavi, "The R-2R and C-2C ladders [a circuit for all seasons]," *IEEE Solid State Circuits Mag.*, vol. 11, no. 3, pp. 10–15, Summer 2019.
- [40] D. Yang, Y. Ding, and S. Huang, "A 65-nm high-frequency low-noise CMOS-based RF SoC technology," *IEEE Trans. Electron Devices*, vol. 57, no. 1, pp. 328–335, Jan. 2010.
- [41] H. Omran, H. Alahmadi, and K. N. Salama, "Matching properties of femtofarad and sub-femtofarad MOM capacitors," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 63, no. 6, pp. 763–772, Jun. 2016.

- [42] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [43] Z. Chen, Q. Jin, Z. Yu, Y. Wang, and K. Yang, "DCT-RAM: A driver-free process-in-memory 8 T SRAM macro with multi-bit charge-domain computation and time-domain quantization," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2022, pp. 1–2.
- [44] Z. Chen et al., "CAP-RAM: A charge-domain in-memory computing 6 T-SRAM for accurate and precision-programmable CNN inference," *IEEE J. Solid-State Circuits*, vol. 56, no. 6, pp. 1924–1935, Jun. 2021.
- [45] H. Jia et al., "Scalable and programmable neural network inference accelerator based on in-memory computing," *IEEE J. Solid-State Circuits*, vol. 57, no. 1, pp. 198–211, Jan. 2022.
- [46] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm² fully-digital computing-in-memory macro supporting wide-range dynamic-voltage-frequency scaling and simultaneous MAC and write operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.
- [47] F. Tu et al., "A 28 nm 29.2 TFLOPS/W BF16 and 36.5 TOPS/W INT8 reconfigurable digital CIM processor with unified FP/INT pipeline and bitwise in-memory booth multiplication for cloud deep learning acceleration," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.
- [48] R. Khaddam-Aljameh et al., "HERMES-core—A 1.59-TOPS/mm² PCM on 14-nm CMOS in-memory compute core using 300-ps/LSB linearized CCO-based ADCs," *IEEE J. Solid-State Circuits*, vol. 57, no. 4, pp. 1027–1038, Apr. 2022.
- [49] P.-C. Wu et al., "A 28 nm 1 Mb time-domain computing-in-memory 6T-SRAM macro with a 6.6 ns latency, 1241 GOPS and 37.01 TOPS/W for 8b-MAC operations for edge-AI devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.



Hechen Wang (Member, IEEE) received the B.S. degree in microelectronics and solid electronics from the University of Electrical Science and Technology of China (UESTC), Chengdu, China, in 2012, and the M.S. and Ph.D. degrees in electrical and computer engineering from Auburn University, Auburn, AL, USA, in 2013 and 2018, respectively, with a focus on the design of integrated digital frequency synthesizer circuits and time-to-digital converters.

In 2014, he was an RFIC Design Engineer with InnoPhase, Inc., Chicago, IL, USA. From 2015 to 2018, he was a Ph.D. Graduate Student Researcher with the Auburn RFIC Design and Testing Laboratory, Auburn University. Since 2018, he has been with Intel Corporation, Hillsboro, OR, USA, where he is currently a Research Scientist with Intel Labs. He has authored or coauthored more than 30 IEEE conference papers and journal articles, U.S. patents, and book chapters. His current research interests include mixed-signal circuits, data converters, digital frequency synthesizers, wireless communication systems, and analog/mixed-signal compute-in-memory for neural networks.

Dr. Wang serving as a member of Technical Program Committee for the IEEE International Symposium on Quality Electronic Design (ISQED). He was a co-recipient of the IEEE Radio Frequency Integrated Circuits Symposium (RFIC) Best Student Paper Award in 2016 and the IEEE Custom Integrated Circuits Conference (CICC) Best Regular Paper Award in 2017.



Renzhi Liu (Member, IEEE) received the Bachelor of Science and Bachelor of Economics degrees from Peking University, Beijing, China, in 2010, and the Ph.D. degree in electrical engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2015.

He was a Post-Doctoral Research Associate with Carnegie Mellon University in 2015. In 2016, he joined Intel Labs, Hillsboro, OR, USA, as a Research Scientist, where he is primarily researching on low-power RF transceivers for emerging applications as well as mixed-signal solutions for in-

memory computing and machine learning applications.

Dr. Liu received the Industry Paper Award 1st Place at IEEE Radio Frequency Integrated Circuits Symposium (RFIC) in 2019 and 2022. He is currently serving as a member of the Technical Program Committee for IEEE International Solid-State Circuits Conference (ISSCC) and IEEE Custom Integrated Circuits Conference (CICC).



Richard Dorrance (Senior Member, IEEE) received the B.S. degree in electrical engineering and computer science from the University of California at Berkeley, Berkeley, CA, USA, in 2009, and the M.S. and Ph.D. degrees in electrical engineering from the University of California at Los Angeles, Los Angeles, CA, USA, in 2011 and 2015, respectively.

In 2015, he joined Intel Labs, Hillsboro, OR, USA, where he is currently a Staff Research Scientist with the RF and Mixed-Signal Circuits Laboratory.

He has authored or coauthored more than 20 publications in leading IEEE conferences and journals, six issued patents, and more than 15 patent applications filed (pending) to date. His current research is largely focused on circuits and systems architectures for machine learning in emerging communications and compute systems.

Dr. Dorrance was a co-recipient of the Best Industry Paper Award at RFIC 2019, Boston, MA, and RFIC 2022, Denver, CO. He is serving as the Chair for the Oregon Chapter of the IEEE Solid-State Circuits Society (SSCS) and an Associate Editor for the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY.



Deepak Dasalukunte (Member, IEEE) received the M.S. and Ph.D. degrees in electrical engineering from Lund University, Lund, Sweden, in 2006 and 2012, respectively.

At Lund University, he was with the Circuits and Systems Group and Center for High-Speed Wireless Communication. In 2010, he was a Visiting Scholar at the Berkeley Wireless Research Center, University of California at Berkeley, Berkeley, CA, USA. From 2012 to 2015, he was with Lantiq Communications India Pvt. Ltd., Bengaluru, India, architecting

next-generation wireline broadband communication products. Since 2015, he has been a Research Scientist at the IP Design and Platforms Group, Intel Labs, Hillsboro, OR, USA. His research interests include hardware architectures for compute and communications systems.

Dr. Dasalukunte has served on the Technical Program Committee of the IEEE Workshop on Signal Processing Systems (SiPS). He is currently the Vice Chair of the IEEE Solid-State Circuits Society (Oregon Chapter).



Dan Lake (Member, IEEE) received the B.S.E.E. degree, the B.S. degree in physics, and the M.S.C.S. degree from Portland State University, Portland, OR, USA, in 1998 and 2007, respectively.

From 1997 to 2007, he was a Software Engineer at Mentor Graphics, Wilsonville, OR, where he developed EDA tools for printed circuit board (PCB) and integrated circuit (IC) flows, including simulation, design for testability (DFT), layout versus schematic (LVS), electrical, and timing verification. Since 2007, he has been with Intel Labs, Intel

Corporation, Hillsboro, OR, USA. He currently manages the Silicon and Systems Prototyping Laboratory, Intel, Hillsboro.



Brent Carlton (Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from Brigham Young University, Provo, UT, USA, in 1999 and 2003, respectively.

Since 2002, he has been with Intel Labs, Intel Corporation, Hillsboro, OR, USA, researching wireless circuits and systems. His expertise includes RF, analog, and mixed-signal circuits, including delta-sigma analog-to-digital conversion, system-level mixed-signal verification, validation, and design automation. He currently manages the Mixed-Signal Circuits and System Team, Intel, Hillsboro, investigating frequency synthesis, radio receiver/transmitter, high-speed data converters, and analog compute-in-memory.